



PRESENTERS



Balmaceda, Sophia Mae Alodia J.- PROJECT LEADER & BACKEND DEVELOPER



Diaz, Dwayne Lester O.- FRONTEND DEVELOPER



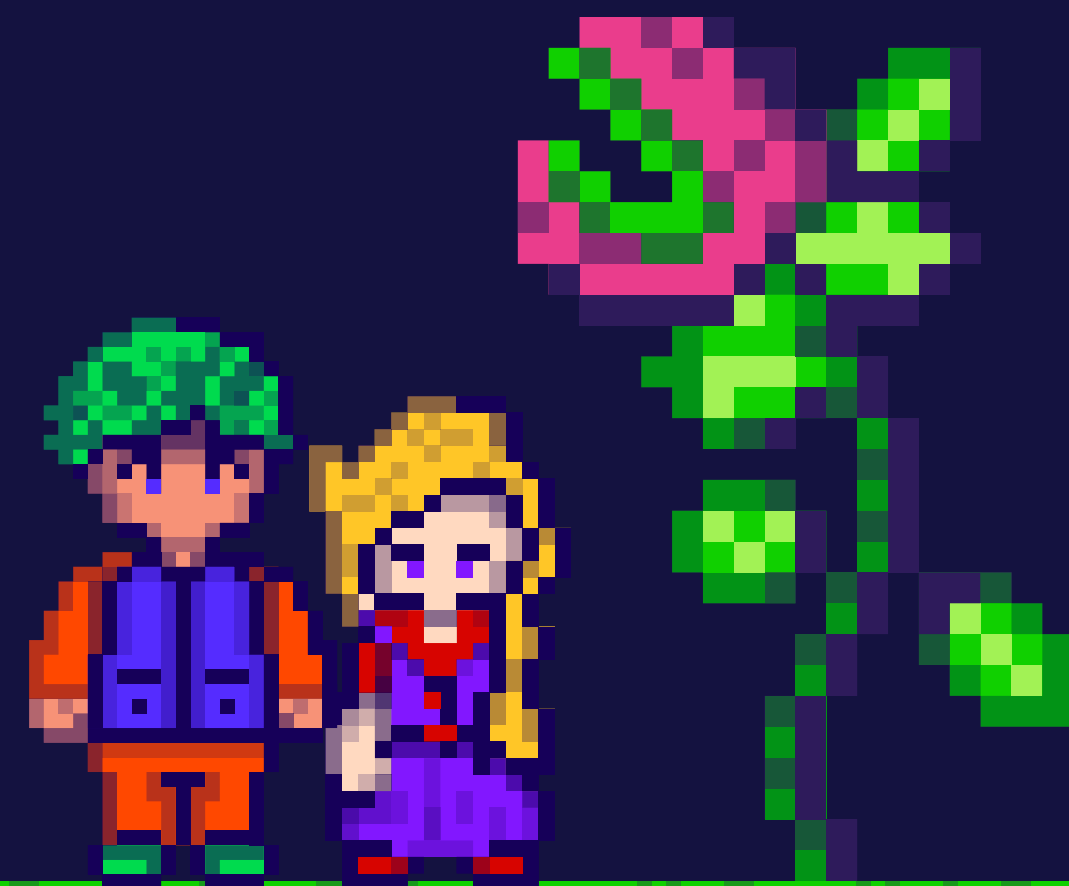
Matutina, Kian Carlo O. - DOCUMENTATION OFFICER



Tuzon, Hafsah A. - GITHUB MANAGER



Velitario, Joshua Cedric B. - DATABASE MANAGER



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COMBAT CODERS

-End of Bugs' Reign-

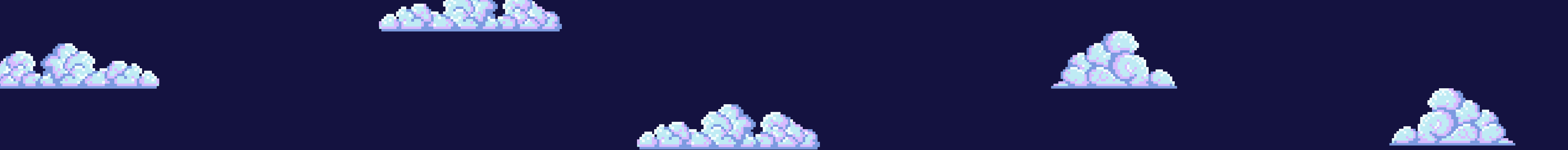




PROJECT INTRODUCTION

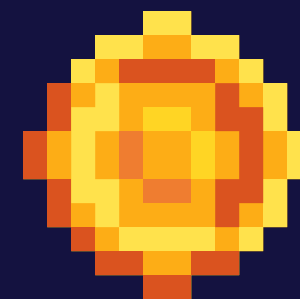
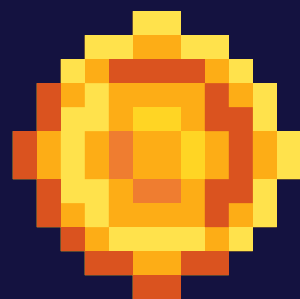
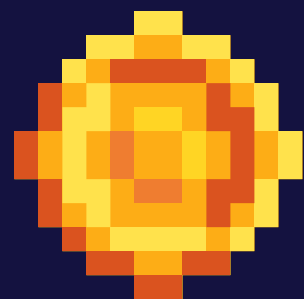


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TO START OFF

Combat Coders is an interactive, game-based learning system designed to improve players' coding skills while enhancing strategic and decision-making abilities. Players navigate using a card-driven system (similar to tarot cards), choosing paths that lead to treasure rooms and battles.



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COMBAT CODERS

HOME

PLAY

LEADERBOARD

LEARN



COMBAT CODERS

The End of Bugs Reign

PLAY



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TECHNOLOGY STACK



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FRONTEND

- **Category** - Technology
- **HTML** - HTML5
- **CSS** - CSS3, Bootstrap 5.3.0
- **JavaScript** - Vanilla JS (ES6+)
- **Fonts** - Google Fonts (Silkscreen, Pixelify Sans, Jersey 10, Tiny5)
- **Icons** - Custom SVG icons
- **Canvas API** - Avatar cropping/export





Backend

- **Category** - Technology
- **Runtime** - Node.js
- **API Base** - <http://localhost:5000/>
- **Endpoints** - REST API
- **Auth** - JWT Bearer tokens
- **Database** - MongoDB

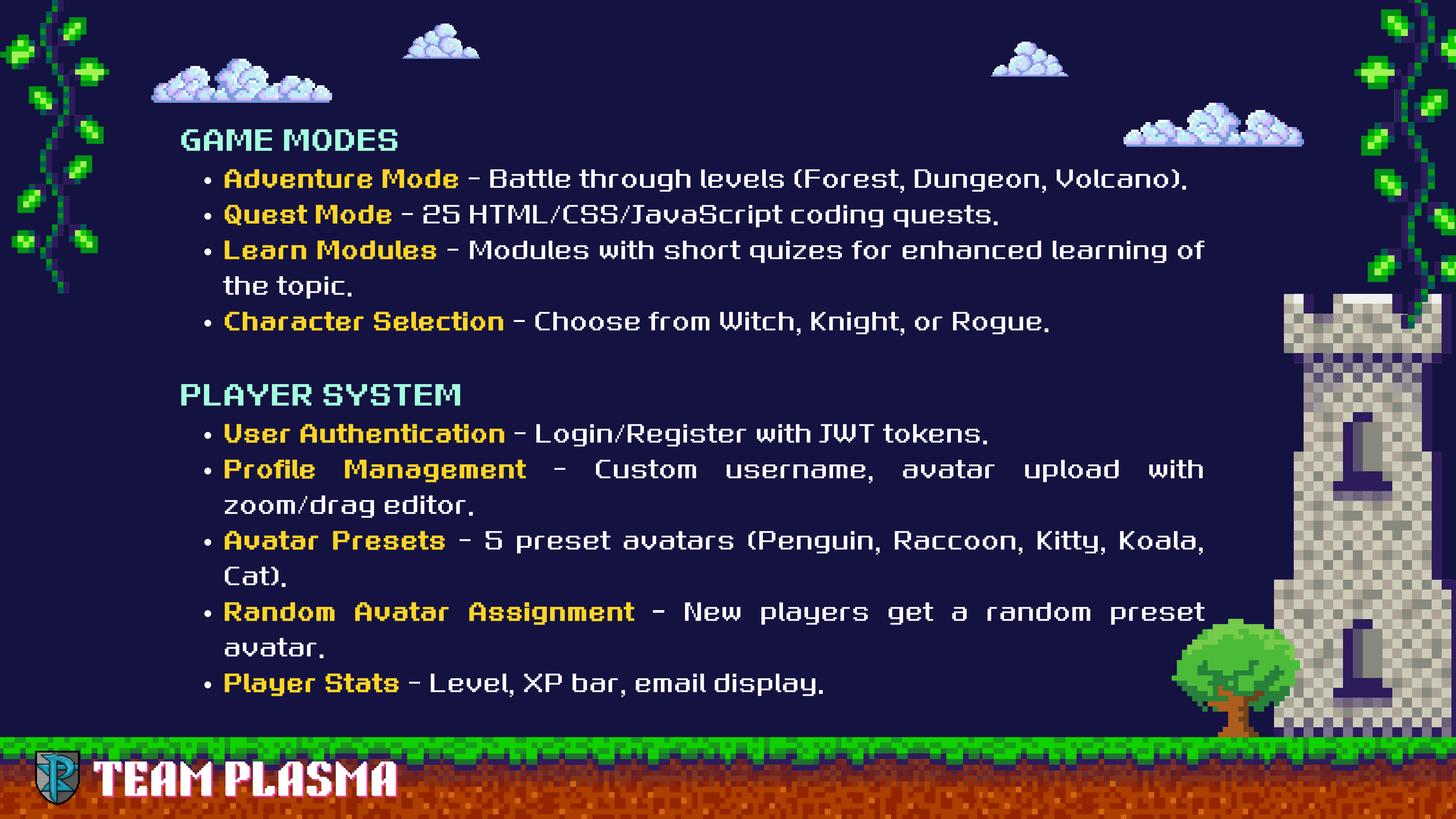




SYSTEM FEATURES OVERVIEW



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GAME MODES

- **Adventure Mode** - Battle through levels (Forest, Dungeon, Volcano).
- **Quest Mode** - 25 HTML/CSS/JavaScript coding quests.
- **Learn Modules** - Modules with short quizzes for enhanced learning of the topic.
- **Character Selection** - Choose from Witch, Knight, or Rogue.

PLAYER SYSTEM

- **User Authentication** - Login/Register with JWT tokens.
- **Profile Management** - Custom username, avatar upload with zoom/drag editor.
- **Avatar Presets** - 5 preset avatars (Penguin, Raccoon, Kitty, Koala, Cat).
- **Random Avatar Assignment** - New players get a random preset avatar.
- **Player Stats** - Level, XP bar, email display.



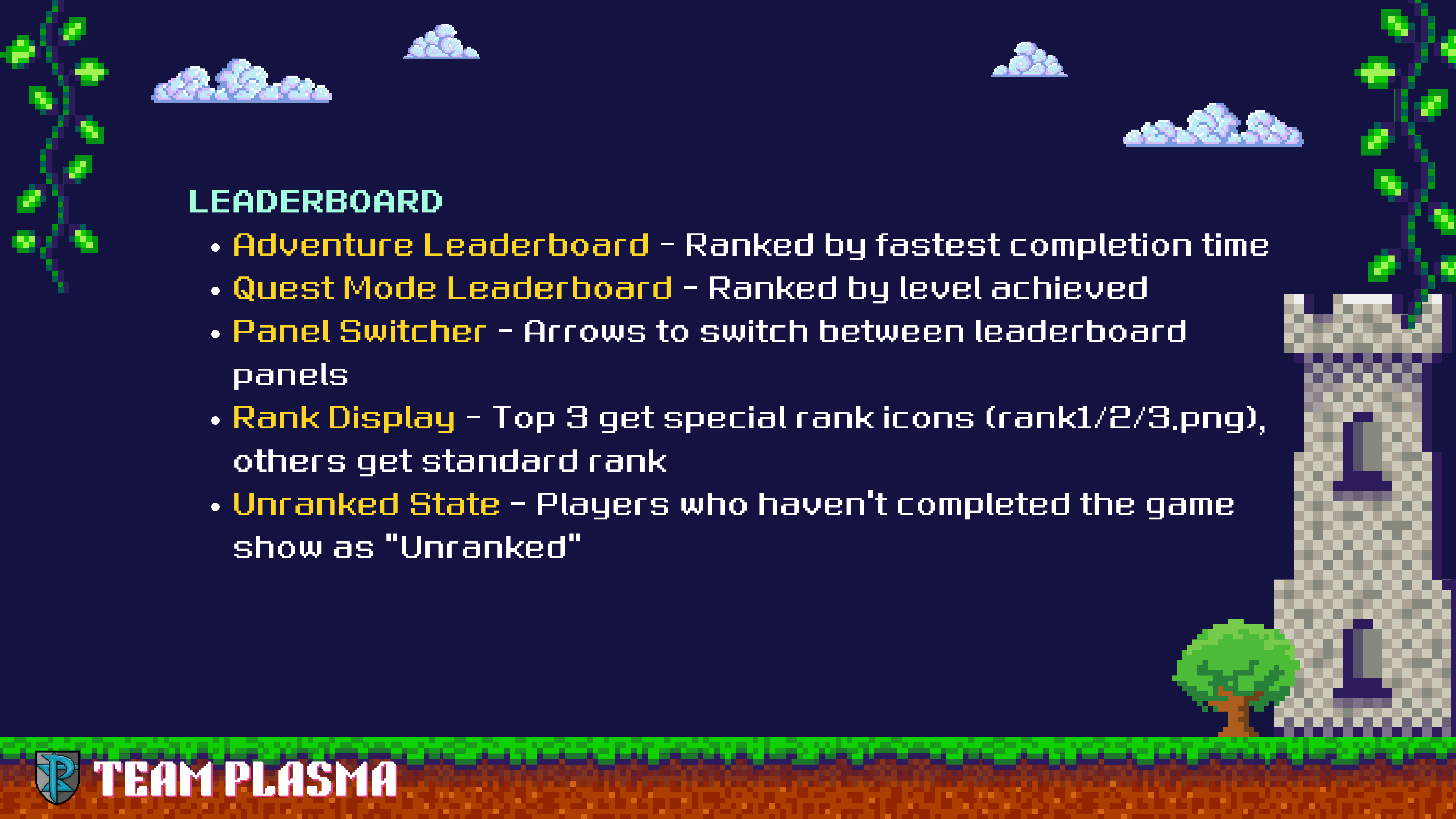


ACHIEVEMENTS & PROGRESS

- **25 Quest Badges** - One badge per completed quest (badge1-badge25)
- **3 Level Medals** - Forest Guardian, Dungeon Conqueror, Volcanic Champion
- **XP System** - Earn XP by completing quests and level.
- **Level Progression** - Level up based on XP earned.



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LEADERBOARD

- **Adventure Leaderboard** - Ranked by fastest completion time
- **Quest Mode Leaderboard** - Ranked by level achieved
- **Panel Switcher** - Arrows to switch between leaderboard panels
- **Rank Display** - Top 3 get special rank icons (rank1/2/3.png), others get standard rank
- **Unranked State** - Players who haven't completed the game show as "Unranked"



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UI/UX FEATURES

- **Pixel Art Style** - Pixelated fonts (Silkscreen, Pixelify Sans)
- **Responsive Design** - Mobile/tablet/desktop layouts
- **Plasma Navbar** - Consistent navigation across all pages
- **Learn Dropdown** - Mechanics and HTML Modules
- **Settings Panel** - Sound controls, change password, delete account
- **Custom Scrollbars** - Styled scrollbars matching the theme
- **Animations** - Leaderboard animations, particle effects on profile





LIVE DEMONSTRATION

WILL BE PRESENTED SHORTLY :)



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CHALLENGES ENCOUNTERED



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CHALLENGES FACED

01

Every time a new feature is added in the battle logic new problems arise such as black overlays and content not properly rendering.

02

During the game, other codes are bleeding into the main code causing bugs.

03

Being committed to iterative visual refinement



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CHALLENGES FACED

04

Scheduling friction: Currently throttled by contributor availability and work-time overlaps.

05

Network Latency: progress being hindered by unstable network infrastructure.

06

Frequent power cutoffs took a toll in our progress.



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FUTURE IMPROVEMENTS



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FUTURE IMPROVEMENTS

01

Gameplay - Introduce new storylines and animations in Adventure Mode

02

Community - Players will be able to see each other's progress and be a much more interactive app.

03

Content - Adding new computer languages (for now it is limited on html, css, and javascript).



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FUTURE IMPROVEMENTS

04

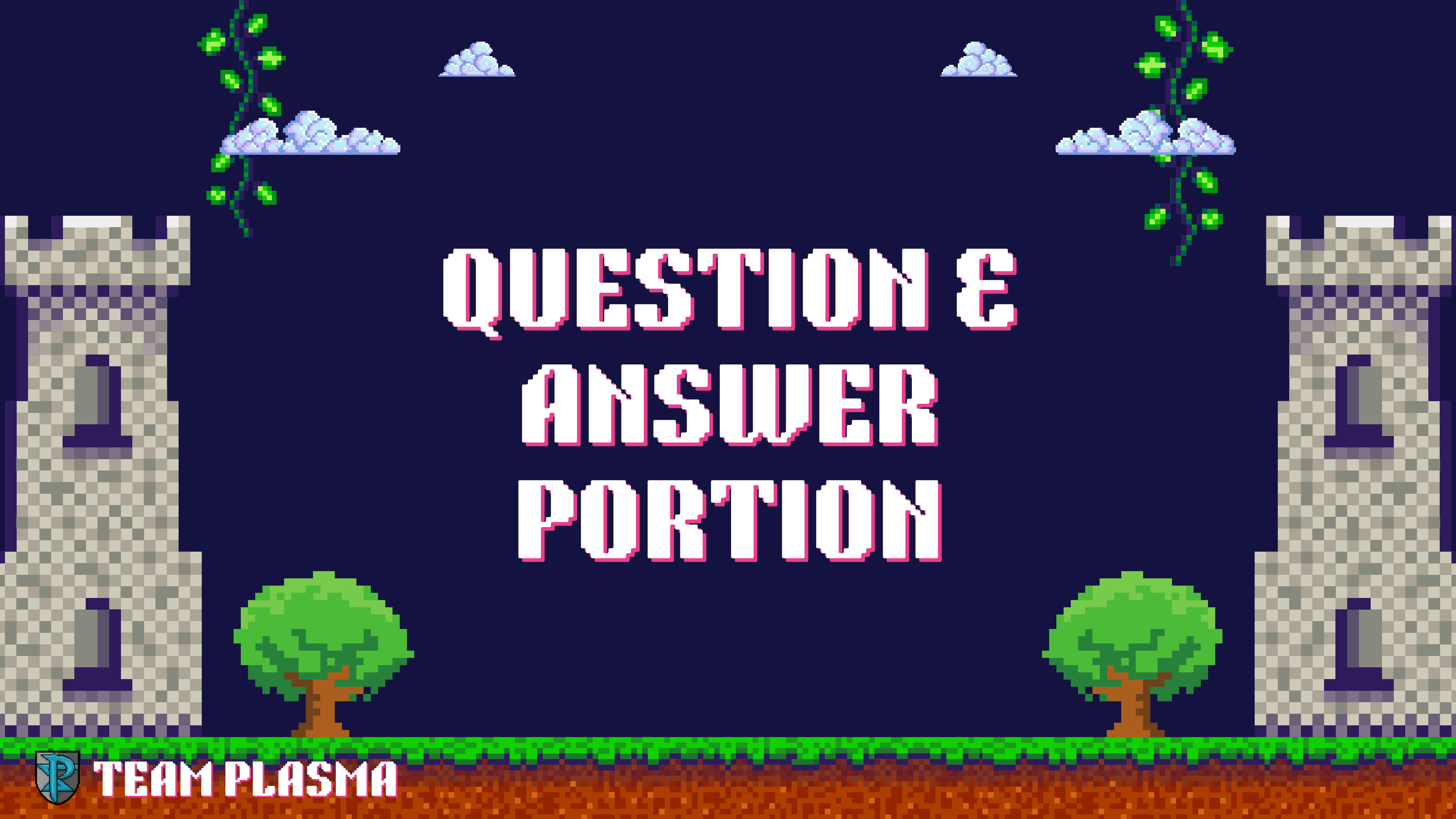
Enhanced UI - Complete UI overhaul for improved cross-platform experience

05

Quality of Life - Implement additional user-friendly features and refinements



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QUESTION & ANSWER PORTION



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THANK
YOU



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